

Research

Enhancing cybersecurity in IoT systems: a hybrid deep learning approach for real-time attack detection

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Abstract

Cybersecurity risks have increased due to the growing ubiquity of Internet of Things (IoT) technology, making attack and anomaly detection a major concern. IoT systems face growing threats from attacks such as Distributed Denial of Service (DDoS), Denial of Service (DoS), Probing, R2L (Remote to Local), U2R (User to Root), Malware, Scanning, Port Scan, and Bot, all of which can seriously jeopardize system integrity, as they become more and more integrated into multiple domains. This paper proposes a hybrid deep learning model (CNN-BiLSTM) using CNN combined with Bidirectional Long Short-Term Memory (Bi-LSTM) for the better detection of such attacks in real-time. The proposed hybrid deep learning model (CNN-BiLSTM) was extensively evaluated on three benchmark datasets, namely KDDCup99, NSL-KDD, and CIC_IDS_2017. The proposed model exhibited outstanding performance on these datasets and yielded an accuracy of 99.9% on KDDCup99, 99.8% on NSL-KDD, and 98.0% on CIC_IDS_2017. Precision, recall, and F1 scores are similar in all attack categories, especially on complex threats such as DoS, DDoS, and malware. A comparison study with the state-of-the-art technique reflects our proposed model's superiority in terms of precision and recall. It offers a good lead toward real-world application and could be piloted in IoT environments by integrating into real-time security platforms to mitigate progressive cyber threats. The contribution of this work is a robust and scalable solution in the fast-growing IoT security area to both present and future challenges in securing critical infrastructures.

Article Highlights

This study presents a novel hybrid deep learning framework that uniquely integrates a Convolutional Neural Network (CNN) for high-dimensional feature extraction with Bidirectional Long Short-Term Memory (BiLSTM) to preserve temporal dependencies in network traffic patterns. Unlike conventional approaches, this model incorporates feature selection using the Least Absolute Shrinkage and Selection Operator (LASSO), ensuring computational efficiency. Furthermore, we employ the Synthetic Minority Over-sampling Technique (SMOTE) to mitigate class imbalance, an issue often overlooked in prior works.

- Hybrid AI model improves cybersecurity: Our approach enhances IoT attack detection by combining CNN and BiLSTM models.
- Tested on real-world datasets: The model is evaluated on benchmark datasets, showing strong performance in detecting threats.
- Supports real-time IoT security: The proposed method is efficient and suitable for real-time cyber threat detection in IoT networks.

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- **Hybrid Deep Learning:** A combination of multiple neural architectures, such as CNN and BiLSTM, to leverage their unique strengths.

Keywords IoT security · Deep learning · IDS · CNN · Bi-LSTM · IoT attacks · Hybrid deep learning

1 Introduction

The Internet of Things (IoT) revolves around linking any device to the Internet [1]. A typical IoT application is structured into three layers: the perception layer, the network layer, and the application layer. The perception layer collects and senses environmental information, which is transmitted to the network layer. The network/transport layer serves as a bridge between the perception layer and the cloud. The application layer then utilizes the data received from the network layer to deliver the necessary operations or services to users [2]. The widespread adoption of the IoT offers numerous opportunities and benefits across various aspects of our lives, including domains such as healthcare, fitness, home energy management, classroom automation, smart cities, and several more [2–4]. The number and diversity of devices have increased, and applications now specifically target individual devices [5]. The world is moving towards the digitization of devices and systems, with nearly all operations relying on the Internet. However, the Internet alone is not enough to support the current digital revolution that is being driven by the development and deployment of new technologies [6]. Recently, the rise of IoT has greatly expanded its usage in communities and services worldwide. In 2017, the number of connected IoT devices reached 27 billion, and this number is expected to grow to approximately 125 billion by 2030 [7]. However, the IoT is also susceptible to numerous vulnerability attacks and anomaly exploits [3]. Simultaneously, addressing cyber-attacks has become increasingly challenging due to the limited resources of many IoT devices constantly evolving nature of network behavior [8], such as storage and computational capacity [4]. Authentication, authorization, and availability are critical components of cybersecurity. Authentication ensures the consistency and integrity of data uploaded to the server; however, replay attacks can compromise this by altering the data. Authorization, or access control, separates legitimate users from illegitimate ones and makes them vulnerable to attacks such as Distributed Denial of Service (DDoS) attacks. These massive DDoS attacks have demonstrated their destructive power by continuously disrupting various infrastructures, causing significant losses, and jeopardizing the overall availability of digital services [1, 9]. Security experts warn of the significant risks posed by IoT devices in various areas [3]. DDoS, Denial of Service (DoS), Probing, Root to Local (R2L), User to Root (U2R), Port Scanning, Malware, and Unauthorized access are all examples of attacks and anomalies that can cause an IoT system to fail [10]. As a result, attacks and anomaly detection have become a growing concern in this sector [3]. Timely identification of IoT attacks is of utmost significance. Recent studies have explored Machine Learning (ML) techniques [3], such as Naive Bayes (NB), Support Vector Machines (SVM), Logistic Regression (LR), and Decision Trees (DT) [3, 10]. Although these old procedures have advantages, they also have significant disadvantages. Extracting valuable insights from them frequently requires substantial human involvement, which might be costly and not feasible. Also, rule-based and supervised learning models are very good at finding known attacks. Still, they struggle to find new, zero-day attacks, especially in the IoT environment, which changes so quickly [11].

Network Intrusion Detection Systems (NIDS) have used deep learning (DL) methods to address these challenges, demonstrating encouraging outcomes. [4]. Researchers have used techniques like Deep Restricted Boltzmann Machines (RBM) and Deep Convolutional Neural Networks (DCNN) to enhance feature extraction and improve classification accuracy. [1]. Techniques like stacked denoising autoencoders (SDAE) have also made DL algorithms better at finding cyber risks in IoT settings. [12]. Nevertheless, the present model's dependence on singular network designs might potentially restrict its capacity to identify correlations across a wide range of variables. Furthermore, the problem of incorrect labeling in training datasets has not been tackled. Small datasets pose challenges for DL models, necessitating the use of strong intrusion detection methods to achieve maximum performance. Precise tuning is also required to optimize performance, and advanced methods are essential for detecting attacks in real time. The primary obstacles are effectively handling feature correlation, resolving inaccurate labels, and exceeding network structure constraints. DL improves security mechanisms for classification and prediction. [9].

IoT Intrusion Detection Systems (IDS) are crucial in protecting networked devices and networks from cyber-attacks. These systems continuously monitor and analyze network traffic and device activity to identify suspicious activities or security breaches [13]. Incorporating real-time warnings and reactions, IDS for the IoT bolsters overall cybersecurity preparedness, guaranteeing the authenticity and reliability of IoT implementations across diverse industries. IDS are categorized into two

main types: host-based, which monitors individual devices, and network-based, which analyzes traffic between devices and the internet, by providing real-time interception of suspicious activities [14]. Traditional intrusion detection systems often face challenges such as low accuracy, ineffective detection, high false-positive rates, and difficulty managing new intrusions. To overcome these challenges, we need to propose a novel deep learning-based approach to identify cybersecurity vulnerabilities and breaches in the IoT environment [15].

The explosive growth in IoT devices has brought new vectors for attacks that are insufficiently covered by conventional security controls. Current IDS tend to fail in identifying sophisticated and dynamic threats in real-time because they have limited capacity for modeling both spatial and temporal relationships between network data. To fill this knowledge gap, we introduce a hybrid deep learning-based IDS model for IoT environments. In contrast to earlier methods based on either CNN or RNN-based models, our approach combines both to utilize their complementary advantages. The objective of this research is to close the research gap by improving the detection accuracy and resilience of IoT IDS solutions using benchmark datasets.

To make our proposed approach clearer, we have expanded our discussion to highlight the key innovations in our methodology. Although CNN and BiLSTM are well-known deep learning models, our hybrid CNN-BiLSTM framework is uniquely structured to leverage spatial feature extraction through CNN and sequential attack pattern recognition using BiLSTM, which enhances the accuracy of detection of evolving IoT threats. Unlike the traditional IDS models, our method combines LASSO-based feature selection for the removal of redundant features and thereby improves classification performance and efficiency in computation. Moreover, SMOTE is applied to the class imbalance inherent in intrusion detection datasets to ensure enhanced detection of minority attack classes, such as R2L and U2R attacks. We further validate the correctness of our proposed system by evaluating our model on benchmark state-of-the-art intrusion detection systems that used the same datasets (KDDCup99, NSL-KDD, CIC-IDS2017) and hence had superior accuracy and recall rates and low false positives. Collectively, all these contributions put our work a great deal toward being the highest advancement for IoT intrusion detection.

Our contributions are as follows:

- Propose a hybrid deep learning model (CNN-BiLSTM) that combines the strengths of CNN and Bi-LSTM for IoT attack detection.
- We conduct comprehensive evaluations on benchmark datasets to validate the model's performance.
- A hybrid (CNN-BiLSTM) model optimized for anomaly detection in network security.
- LASSO-based feature selection for reducing dimensionality and computational overhead.
- Handling of dataset imbalance using SMOTE for robust classification.
- Extensive benchmarking against existing methods, demonstrating superior accuracy across multiple datasets.
- The use of the proposed hybrid deep learning model (CNN-BiLSTM) is explored for detecting various types of attacks, including DDoS, Port Scan, Web, Bot, Scanning, Unauthorized, Probe, R2L, U2R, and DoS, within IoT environments.
- Evaluate the hybrid model (CNN-BiLSTM) on three diverse datasets, namely KDDCup99, NSL-KDD, and CIC_IDS_2017, demonstrating its effectiveness in various scenarios.
- Our model significantly outperforms several state-of-the-art approaches in terms of accuracy, precision, recall, and F1-score.
- Achieve high accuracy rates on all datasets, indicating the proposed hybrid deep model (CNN-BiLSTM)'s potential for real-world deployment.

The rest of this paper is structured as follows: Sect. 2 reviews the related work. Section 3 presents the proposed methodology, covering data distribution, feature selection, data balancing, and the hybrid model. Section 4 details the empirical evaluation, including the experimental setup and result analysis. Finally, Sects. 5 and 6 offer the conclusion and future directions, and Future recommendations.

2 Related work

This section contains a selection of studies based on several criteria. First, we preferred studies that focused on IDS for IoT networks and traditional network security threats. We excluded the studies related to fraud detection and general anomaly detection, which are unrelated to the domain. Then we selected papers using deep learning algorithms (CNN, LSTM, BiLSTM, and hybrid models) or machine learning classifiers (RF, SVM, XGBoost, etc.) for intrusion detection, eliminating purely statistical or rule-based IDS methods. Also, we considered studies using open-source datasets of IoT such

as KDDCup99, NSL-KDD, CIC-IDS2017, and Bot-IoT and excluded proprietary ones due to concerns about reproducibility. Finally, we considered the studies reporting detailed performance metrics like accuracy, precision, recall, F1-score, and false positive rate (FPR), and excluded those without empirical verification or comparative results. A considerable amount of effort has been dedicated to the development of intrusion detection systems using a wide array of ML and DL models. For example, in the study [1] The author developed a smart city platform to detect replay and DDoS attacks using authentic smart city datasets. The model architecture includes an input layer, a deep CNN with seven hidden layers, a Global Average Pooling (GAP) layer, a deep Restricted Boltzmann Machine (RBM) model with two hidden layers, and a SoftMax output layer. The Deep RBM model learns high-level features from raw data, while the GAP layer enhances feature maps and serves as regularizer. The SoftMax layer completes the model with a hybrid algorithm that alternates between training CNN and RBM, making structural updates, and iteratively evaluating performance for increased accuracy.

This study was presented in the [2], showed how to use network data to find intruders using generative deep learning methods, namely adversarial autoencoders and bidirectional generative adversarial networks (BiGAN). The adversarial autoencoder model generates a latent representation $E(x)$ of size 6 from the input, as well as reconstructs the original data. A discriminator evaluates the generated features, and a random variable distinguishes between them. For intrusion detection, test data is encoded into the latent space, and a classifier like K-Nearest Neighbor (KNN) is trained with this information. It was trained in over 1000 iterations with a size of 10 and a learning rate of $10^{(-4)}$. The Adam optimizer and mean square error were used as the loss functions.

The author [3], has proposed a novel ensemble hyper-tuned model has been proposed for the automatic and effective detection of attacks and anomalies in IoT sensors. This robust model integrates an ensemble machine learning method with feature selection to enhance detection performance. The Distributed Smart Space Orchestration System (DS2OS) used 357,952 samples with 13 features. The process begins with feature selection to reduce the dataset size and improve anomaly detection. They then apply a hyper-tuned gradient-boosting ensemble algorithm. The model involves data preprocessing, converting data into feature vectors, optimizing hyperparameters, and evaluating the model using various metrics to improve attack and anomaly detection.

In the study presented by [4] has designed and implemented a lightweight ML-based IDS on a Raspberry Pi using the correlated-set thresholding on the gain-ratio (CST-GR) algorithm. The CST-GR algorithm detects attacks by selecting the set of features that are most strongly related using the CFS algorithm's merit function. This function figures out the gain ratio for each feature so that it can pick the most important ones and leave out the least important ones. This approach maintains predictive model accuracy. The proposed IDS, tested on the Bot-IoT dataset, showed effective performance with reduced processing time and maintained accuracy.

In the study presented by [5], has introduced a distributed framework, Adept, to distinguish between different phases of coordinated attacks. Adept operates in three stages: initially, it monitors IoT device network traffic for anomalies, alerting a security manager if detected. The security manager uses frequent item mining (FIM) to analyze these alerts for correlated patterns over time and space. Lastly, ML techniques, including k-NN, RF, and SVM, classify specific attack stages using alert-level and pattern-level information as features. Adept effectively balances privacy with robust attack detection and classification.

The study presented by [7] looked at an ML-based attack and anomaly detection method that uses algorithms like LR, SVM, DT, RF, ANN, and KNN to help protect smart cities from cyber threats. This author explored ensemble techniques like bagging, boosting, and stacking to enhance detection performance. The study also integrated feature selection, cross-validation, and multi-class classification to improve the efficacy of the system.

To efficiently detect and mitigate various types of DDoS attacks in heterogeneous IoT infrastructures through binary and multi-class classification, the author [9] Introduced the HetIoT-CNN IDS, a DL-based IDS. The CNN architecture consists of two 1D convolution layers, one 1D max-pooling layer, and a fully connected dense output layer. They employed two dropout layers for binary classification. Each model layers do a different job: convolution layers use stride and padding to extract linear features; max pooling extracts the most activated features to cut down on computation time and errors; and dropout layers with a 0.5 value stop the model from fitting too well. The flattened layer converts multi-dimensional data into a one-dimensional format for the fully connected dense layer, which uses the SoftMax activation function to identify and classify DDoS attacks. Their architecture ensures efficient detection and classification within the IoT infrastructure.

In the study presented by [10], compared the performance of several ML models to accurately forecast attacks and anomalies in IoT systems. The algorithms evaluated included Artificial Neural Networks (ANN), DT, RF, SVM, and LR. Utilizing the DS2OS dataset.

In the study presented by [11], introduces a DL technique for IoT IDS using an asymmetric logarithmic loss function and a CNN. This model addresses sustainability and practicality issues and outperforms traditional methods in detecting

IoT attacks. Feature fusion is used to represent a feature vector and its classification label, with input and output vectors from an unknown joint distribution forming the dataset. Convolution layers translate input vectors into output vectors, with weight sharing reducing training complexity. Feature fusion enhances semantic information and detail perception, training a classifier with these features. The CNN model is optimized to minimize the loss function, measuring the discrepancy between expected and actual outcomes, using gradient descent methods.

In the study presented by [12] Investigates enhancing IoT fusion security through a DL-denoising autoencoder. A hierarchical intrusion detection model, termed Stacked Denoising Autoencoder Supporting Vector Machine (SDAE-SVM), is constructed using a three-layer neural network of self-encoders. The model employs layer-by-layer pretraining and fine-tuning to obtain sample data after dimension reduction. Input data is encoded into a compact representation through input, hidden, and output layers by an autoencoder to minimize the output. Activation functions such as Sigmoid or Tanh, along with weights and biases, are applied during the encoding and decoding processes. A trained denoising autoencoder (DAE) enhances robustness and data recovery by efficiently removing noise from input data. An SDAE is formed by stacking multiple DAE, where each DAE uses the output of the previous one as an input to learn increasingly complex features. Gradient descent updates the model parameters (weights and biases) during training.

In the study presented by [14], introduces a method for detecting injection attacks in IoT applications, particularly in smart cities. The approach uses two feature selection techniques, constant removal and recursive feature elimination, evaluated across DL classifiers such as SVM, RF, and DT. These techniques reduce the critical features needed to detect injection attacks from 13 to 8 using preprocessed datasets. Constant removal, a variance-based feature selection method, ranks and selects the top N features while eliminating zero-mean features and columns with minimal variance differences, effectively distinguishing injection attack classes from normal classes. Recursive feature elimination, a wrapper-based technique, removes the lowest-weighted features using a search mechanism, yielding more accurate results than filter-based techniques. The accuracy of the selected features was evaluated using a T-test with the proposed approach.

In the study presented by [15], an AI-supported advanced intrusion detection framework was proposed to identify malicious cyberattacks, enhancing IDS by leveraging Deep Neural Network (DNN), Recurrent Neural Networks (RNN), and Multilayer Perceptrons (MLP). IoT-driven industrial IoT (IIoT) networks use GAN to detect cyber threats. The framework includes a DNN with one input layer, three hidden layers, and an output layer. Tested on the NSL-KDD dataset.

In the study presented by [16], has presented a novel adversarial attack on DL-based NIDS in IoT environments has been presented. The method uses model extraction and saliency map analysis to create an exact copy of the DL model with very little training data and examples that are not what the model expects. This study utilizes the Kitsune NIDS, which features five unsupervised learning modules—Packet Capturer, Packet Parser, Feature Extractor (FE), Feature Mapper (FM), and Anomaly Detector (AD). The FE module employs "damped incremental statistics" for high-speed feature extraction, while the FM module uses agglomerative hierarchical clustering to group features. The AD module leverages auto-encoders and normalized Root Mean Square Error (RMSE) values for anomaly detection.

In the study presented by [17], a two-stage ensemble deep learning framework aimed at detecting and attributing attacks within imbalanced Industrial Control Systems (ICS) datasets. The method employs one-vs.-all classifiers for attribute attribution and an ensemble representation learning model for attack detection. A special DNN model with two stacked autoencoders and encoder and decoder layers is used to map inputs to higher dimensions and reconstruct input representations by repeatedly reconstructing them. The framework enhances performance by integrating two autoencoders: Principal Components Analysis (PCA) for dimensionality reduction, a DT classifier, and a one-class SVM to identify unseen attacks.

In the study, the author [18] proposes a novel method for identifying FDI attacks using Autoencoders (AE). This technique leverages the correlation between sensor readings in both space and time to detect manipulated data. Denoising Autoencoders (DAE) are employed to clean the manipulated data. The AE architecture consists of three layers: input, hidden, and output, with training conducted via backpropagation and the Mean Squared Error as a common error metric. AE identifies false data injection attacks by disrupting the correlation model and learning the relationships and data structure.

In the study presented by [19], proposes a DNN for intrusion detection in Message Queuing Telemetry Transport (MQTT-based) protocols, comparing its performance with classical ML algorithms such as NB, RF, k-NN, DT, LSTM, and GRU. The DNN architecture consists of an input layer, two interconnected hidden layers, and an output layer. Data is processed from the input layer through the hidden layers, with the output layer adapting to the classification tasks. The model uses sigmoid activation for binary classification, SoftMax for multi-classification, and Rectified Linear Unit (ReLU) activation in the hidden layers, tailored specifically for networks using the MQTT protocol.

In the study presented by [20] Specifically focuses on employing ML to improve the identification of cyber threats, especially False Data Injection Attack (FDIA) attacks on smart grid systems, where accurate state estimation and detection of malicious meters are involved.

2.1 Limitations and potential challenges associated with attacks

Despite the vast improvement in the detection of attacks through various advanced systems, some limitations and challenges have been found in previous studies. [1] Models suffer from a data imbalance issue where minority classes of attacks cannot be detected; thus, learning becomes biased and less accurate in detecting less frequent attack types. [2] Deep learning-based models may pose a computational complexity problem in real-time detection, which complicates their deployment in resource-constrained environments like IoT devices. [3] Explainability in AI-driven models limits their practical adoption into real security threats, as security analysts rely more on interpretable decisions to adequately respond to threats. [4] Relying on predefined attacks limits the adaptability of the systems to novel and evolving cyber-attacks. [5] Some of the detection methods operate with outdated datasets that are not representative of modern attack vectors, further limiting their efficacy in a real-world application. [6] The trade-off between detection performance and false positive rate is still one of the issues since high false alarms can overload security teams and decrease trust in automated systems. [7] Robust feature selection and preprocessing techniques must be used since noisy and redundant data can easily degrade model performance. [8] Many existing studies were not validated against multiple datasets. [9] Finally, attack detection models are vulnerable to adversarial manipulations, and attackers can craft inputs to evade detection. Addressing these limitations is essential to developing more effective and resilient attack detection systems.

Key Limitations in Existing Studies: Our review of related work, summarized in Table 1, revealed several critical gaps in existing intrusion detection systems for IoT security.

3 Proposed methodology

In this section, we propose a hybrid deep learning model (CNN-BiLSTM) that integrates several distinct processes. As shown in Fig. 1, the model begins with dataset collection and examination. Following this, data preprocessing is performed, which includes cleaning, visualization, feature selection using LASSO, data balancing using SMOTE, and vectorization. These steps transform the data into feature vectors. The dataset is then divided into training and testing sets in an 80–20 ratio [21]. The training set is used to train the proposed hybrid deep learning model (CNN-BiLSTM), while the final hybrid deep learning model is assessed against the testing set using various evaluation metrics.

3.1 Description of the dataset

For our research on IoT attack detection using a hybrid deep learning model (CNN-BiLSTM), we utilized three benchmark datasets: CIC_IDS_2017, NSL_KDD [22], and KDDCup_99. The CIC_IDS_2017 dataset, provided by the Canadian Institute for Cybersecurity, includes a comprehensive range of modern attacks and benign traffic, offering a realistic representation of contemporary network activities [23, 24]. The NSL_KDD dataset is an enhanced version of the original KDD Cup 1999 dataset [25]. Finally, the KDDCup_99 dataset, despite its age, remains a foundational resource for network intrusion detection research due to its extensive collection of network traffic data and diverse attack types. By leveraging these datasets, we aim to create a robust and generalizable deep learning model capable of effectively identifying and mitigating cyber threats in IoT environments. The data sources were available in multiple formats, including original files with a .Pcap extension and comma-separated values (CSV) files. Pcap files, typically used by Wireshark programs, contain data packets from a network and are commonly used for analyzing network data characteristics. These files are organized based on attack categories and subcategories to aid in labeling. While datasets such as KDDCup99 and NSL-KDD originate from traditional network security environments, they contain many of the same attack types observed in IoT networks. These datasets have been widely used in IoT security research due to their comprehensive attack taxonomies, making them valuable for evaluating intrusion detection models for IoT applications. Table 2 summarizes the number of attacks present in the dataset. The different classes of attacks in the dataset to target an IoT environment are overviewed in Table 3.

Table 1 Key limitations of existing studies

Limitations	Existing Works	How Our Model Addresses it
High False Positive Rates	Studies in [6] and [9] report high detection accuracy but do not effectively reduce false alarms	Our model integrates LASSO-based feature selection to improve decision boundaries and reduce false positives
Lack of Generalization Across Attack Types	IDS models in [1, 5], and [8] perform well on specific attack categories (e.g., DoS) but struggle with minority attack types such as R2L and U2R	We apply SMOTE oversampling to improve the detection of under-represented attacks, enhancing generalization
Dataset Constraints & Limited IoT-Specific Data	Studies in [2] and [10] rely on traditional IDS datasets that may not fully capture IoT-specific attack patterns	While we use KDDCup99, NSL-KDD, and CIC-IDS2017, we acknowledge this limitation and propose future validation with TON_IoT and Bot-IoT datasets

Fig. 1 The flowchart illustrates the application of a proposed hybrid model (CNN-BiLSTM) on three IoT datasets

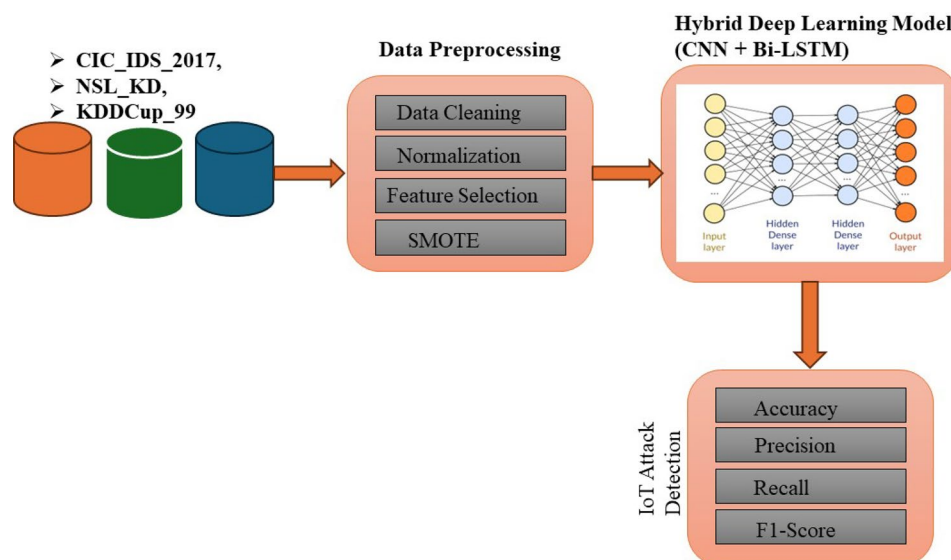


Table 2 IoT attack summary of a dataset

CIC_IDS_2017		KDDCup_99		NSL_KDD	
Attacks	Frequency Counts	Attacks	Frequency Counts	Attacks	Frequency Counts
Bening	2,094,484	Normal	812,813	Normal	812,528
DDoS	128,014	DoS	247,042	DoS	199,245
DoS	193,745	Malware	6228	Probe	8583
Port Scan	90,694	Scanning	8841	R2L	6173
Web	11,340	Unauthorized	43	U2R	43
Bot	1948	Web	24	Web	24

3.2 Feature selection using LASSO

A regularization requirement is introduced by the robust regression approach LASSO, which improves model performance and interpretability. LASSO was preferred for feature selection in this study because it offers the advantages of feature selection and regularization all at once, making it perfect for high-dimensional cybersecurity data. The use of L1 regularization for LASSO places emphasis on sparsity through the automatic elimination of irrelevant features and retaining the most informative features, which benefits model interpretability and generally reduces overfitting. It is computationally efficient compared to techniques such as Recursive Feature Elimination (RFE) and supports faster training and inference, critical for real-time IoT intrusion detection. PCA, which results in loss of feature interpretability, or Mutual Information, which retains redundant features, LASSO effectively selects the most relevant attributes while keeping the model efficient. By applying an L1 penalty on the coefficient magnitudes, LASSO promotes sparsity in the model instead of conventional linear regression, which may incorporate all characteristics regardless of their importance. This penalization performs automatic feature selection by shrinking some coefficients to zero. Mathematically, the objective function for LASSO is given below equation:

$$\min_{\beta_0, \beta} \left\{ \frac{1}{2n} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p X_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Table 3 Overview of attacks present in the dataset

S. No	Attacks	Descriptions
1	DDoS	A DDoS attack involves flooding a target system with traffic from many sources, making the service unavailable to authorized users [23, 26]
2	DoS	A DoS attack disrupts the normal functioning of a service by overwhelming it with excessive requests, causing it to crash or become unavailable [21]
3	Port Scan	A port scan is performed to find open ports and services on a network. This is frequently done as a first step towards more focused attacks
4	Web	These assaults take advantage of online application flaws like SQL injection and cross-site scripting to alter or obtain unauthorized access to data [21]
5	Bot	Bots are automated scripts that carry out repetitive operations. They are frequently used maliciously for DDoS, spamming, and credential-stuffing attacks [21]
6	Malware	Malware spreads via infected files or websites and is harmful software intended to enter a system, cause harm, or steal data
7	Scanning	Scanning is the systematic practice of inspecting computer systems for vulnerabilities or open ports. It is widely used to identify potential exploit targets
8	Unauthorized	This phenomenon arises when a malevolent individual obtains entry to a system or network without appropriate authorization, frequently resulting in the pilferage of data or the disturbance of operations
9	Probe	A probe is an act of survey in which attackers collect information about a networks defenses and vulnerabilities to strategize subsequent attacks [21]
10	U2R	U2R attacks involve exploiting vulnerabilities to escalate privileges from a normal user to root access, gaining complete control over the system [21]
11	R2L	R2L assaults occur when an external assailant obtains local access to a system, typically by employing methods such as phishing or exploiting remote services [22]

where:

y_i is the target variable.

X_{ij} represent the feature values.

β_j are the coefficients to be learned.

λ is the regularization parameter controlling the degree of sparsity.

With high-dimensional datasets—where the number of features may exceed the number of observations—LASSO excels in handling large amounts of data. LASSO enhances model accuracy by decreasing overfitting and simplifies the model, making it easier to understand and more computationally economical by keeping only the most significant predictors [13, 24]. We employed a cross-validation technique to optimize the choice of λ and ensure the model performs well when applied to new data ($cv=5$). Following pipeline fitting, features with coefficients that are not zero are chosen, and those choices shrink the dataset. This reduction of dimensionality improves the interpretability of the model and computational performance. LASSO is represented in Algorithm 1.

Algorithm 1. LASSO [31]

Input: Processed dataset X' and target variable Y

Output: Selected Features F

- i. Lasso Regularization:
Apply Lasso on X' and Y
- ii. Feature Selection:

$$F = \{j \mid \hat{\beta}_j \neq 0, \text{ for } j = 1, 2, 3, \dots, p\}$$

(Select a feature with a non-zero coefficient)

- iii. Update Feature Set:

$$F \leftarrow F \cup \{\text{selected features}\}$$

- iv. Return Selected Features:
Output: F
-

3.3 SMOTE for balancing multiclass classification dataset

To solve the class imbalance issue arising from the multiclass classification of IoT attack detection, we utilized the SMOTE. SMOTE is a popular technique that uses interpolation between existing examples to create synthetic samples for minority classes. SMOTE was applied independently to every minority class in the sample in the context of multiclass categorization. Synthetic examples were produced for each minority class by choosing a sample and its closest neighbors and then creating new samples along the line segments that joined the selected sample to its neighbors. By adding synthetic data to the dataset, this procedure efficiently balances the class distribution and enhances the hybrid deep learning models' capacity to learn from all classes equally, as represented in Fig. 2.

SMOTE improves the hybrid deep learning model's effectiveness in identifying less common attack types by reducing the bias toward the majority classes, which helps to produce more accurate and dependable IoT attack detection [27]. The data balancing process using SMOTE is illustrated in Algorithm 2.

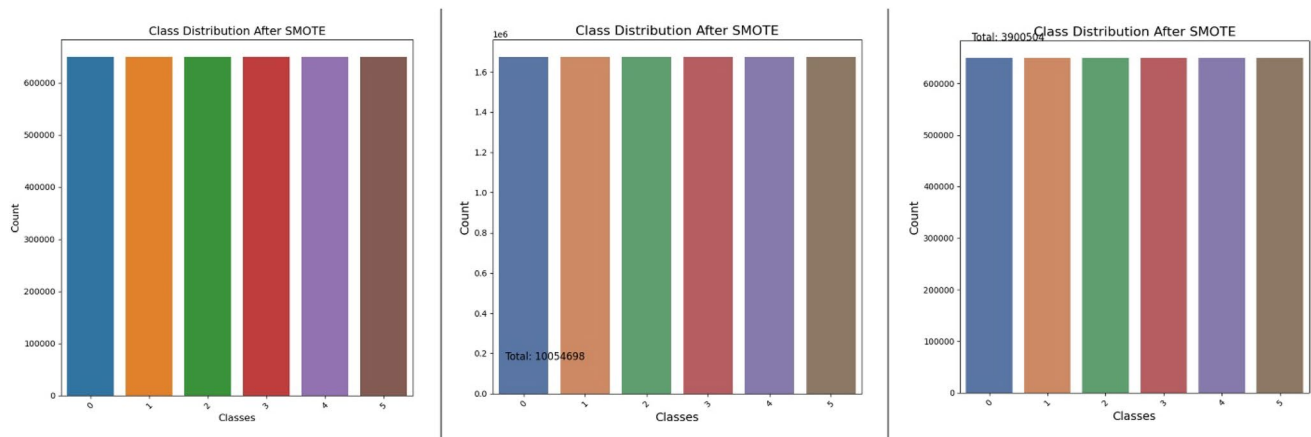


Fig. 2 Illustrates the data distribution after applying SMOTE for datasets NSL_KDD, CIC_IDS_2017, and KDDCup99, respectively

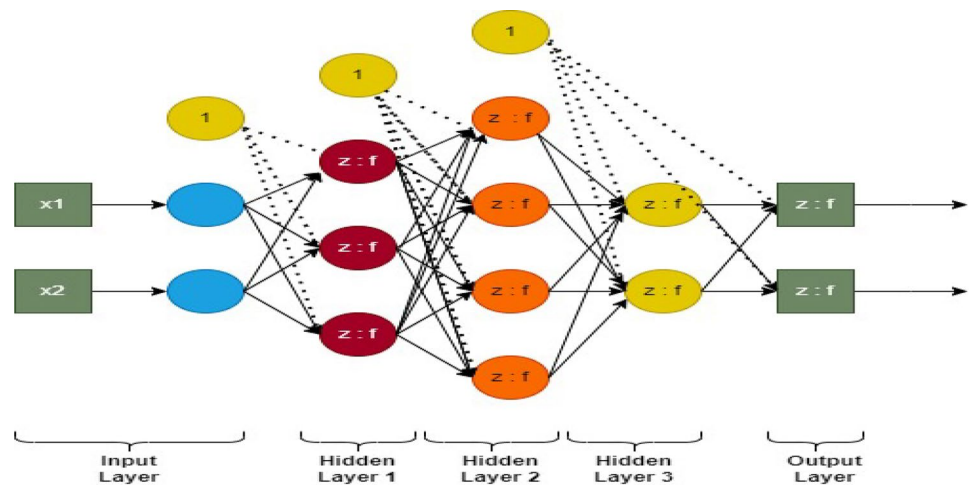
Algorithm 2.

-
- i. **Input:**
 $D = \{(x_i, y_i)\}_{i=1}^n$ where x_i is the feature vector and y_i is the class label.
 $N_{minority}$ = number of samples in the minority class.
 k = number of nearest neighbors.
 T = number of synthetic samples to generate.
 - ii. For each minority class C_j :
 Identify the samples $X_j = \{x_i | y_i = C_j\}$
 For each x_i in X_j :
 Find k -nearest neighbors of x_i within X_j .
 Randomly select one of the k -nearest neighbors of x_{nn} .
 Generate synthetic sample:
 $\text{synthetic_sample} = x_i + \delta \times (x_{nn} - x_i)$
 Where $\delta \sim U(0, 1)$
 - iii. Repeat step ii until T synthetic samples are generated for each minority class.
 - iv. Combine the original dataset D with the generated synthetic samples to form a new balanced dataset.
 - v. **Output:** Balanced dataset D'
-

3.3.1 Hybrid deep model (CNN + Bi-LSTM)

A combination of multiple deep learning models that work together to extract richer feature representations from the data. To capture the spatial and temporal aspects of IoT network traffic, we propose a hybrid deep learning architecture that incorporates CNN and Bidirectional Long Short-Term Memory networks [28, 29]. CNN is utilized for its strength to extract high-level spatial features from input data, whereas BiLSTM can capture temporal dependencies in forward and backward directions, which is important for discovering sequential patterns of network traffic. This integration provides a solid architecture for detecting known and emerging attack behaviors. The model is suitable for deployment in real-world environments such as smart city infrastructure, healthcare monitoring systems, and smart industrial setups. By integrating it with an SDN-based controller or edge computing node, real-time response to intrusion attempts can be achieved with minimal latency. The choice of hybrid CNN-BiLSTM architecture is based on the complementary advantages of both models in processing network traffic data, which tends to have both spatial and temporal dependencies. CNN

Fig. 3 A general architecture of the CNN



are particularly efficient at extracting local and hierarchical spatial features from input sequences and hence are appropriate for detecting patterns in packet structures or flows. Conversely, BiLSTM networks can learn long-term temporal dependencies both in forward and backward directions, which is important for comprehending sequential activity in network traffic, which could signify multi-step or dynamic attacks. With these two architectures combined, the model can utilize CNN feature extraction ability with BiLSTM awareness of time to present a more complete view of the data and increase detection accuracy. The use of LASSO for feature selection guarantees dimensionality reduction and removal of redundant or less important features, thus minimizing overfitting and promoting interpretability of the model. SMOTE is also employed to address the issue of class imbalance, which is common in intrusion detection data, thus enhancing the performance of the model on minority class attacks like U2R and R2L. Nonetheless, there are some potential drawbacks: higher computational cost because of the hybrid architecture, which might not be ideal for real-time operation on resource-limited IoT edge devices. Also, the need for labeled data for supervised learning is a challenge, as it can be hard to get high-quality, current attack data. Additionally, while SMOTE assists in balancing the dataset, it can add artificial samples that might not represent real-world behavior as accurately as desired. Subsequent work will involve optimizing the architecture for low-latency settings, investigating the use of unsupervised or semi-supervised learning, and deploying methods such as federated learning to provide privacy and scalability in distributed IoT systems.

3.3.2 CNN layer

The CNN, also known as ConvNet, is characterized by a deep and hierarchical architecture designed for high computational efficiency when processing complex data [30]. In deep learning, CNNs typically represent data through cascading layers of linear and nonlinear operations. Unlike traditional methods, CNN relies on independent features and requires minimal preprocessing, reducing the need for prior knowledge [9]. The fundamental components of a CNN are its ConvNet layers, which bear the computational load and facilitate the transfer of data from one layer to the next [22]. CNN uses a modified design of the multilayer perceptron, emphasizing local connectivity and weight sharing to minimize processing [31]. For self-learning, the CNN layers compute the dot product of the data represented in matrix form, with the learning process driven by learnable parameters and various kernels [30]. CNN comprises three types of layers: input, hidden, and output. The hidden layers employ various filters, such as kernels, max pooling, and min pooling, along with fully connected layers. As data moves through the network, it traverses each forward or backward layer. The kernel, a crucial element in the network, slides across the input with a specific height and width [15]. The general architecture of the CNN is depicted in Fig. 3.

3.3.3 Input layer

The input data is reshaped to align with the Conv1D layers format, treating each feature as a distinct channel, and ensuring compatibility with the model architecture mathematically. $X \in \mathbb{R}^{n \times 1}$ denotes the input sequence, where n = number of features. Each input sample is a column vector ready for the following layers to process [21].

3.3.4 Convolutional layer

The proposed hybrid deep model employs two Conv1D layers, with kernel sizes of 3 and filter sizes of 32 and 64, respectively [22]. Each convolutional operation is followed by Batch Normalization and LeakyReLU activation.

$$f(y) = \begin{cases} y & \text{if } y > 0 \\ 0.1y & \text{if } y \leq 0 \end{cases}$$

which enhances training stability and speed, and introduces necessary non-linearity. The convolution process for every position i The sequence is as follows:

$$y_i = \sum_{k=1}^3 w_k x_{i+k-2} + b$$

w_k denotes the filter weights and b represents the bias term.

3.3.5 Batch normalization layer

Normalizes the convolutional layers output to have zero mean and unit variance, accelerating training and providing regularization represented in the equations.

$$\hat{y} = \gamma \frac{y - \mu}{\sigma} + \beta$$

where μ and σ are the batch mean and variance, and γ and β These are the learnable parameters.

3.3.6 Pooling layer

MaxPooling1D layers, with a pool size of 2, are implemented to down-sample the feature maps, effectively reducing computational complexity and mitigating the risk of overfitting, which is represented in the equations.

$$z_i = \max(y_{2i-1}, y_{2i})$$

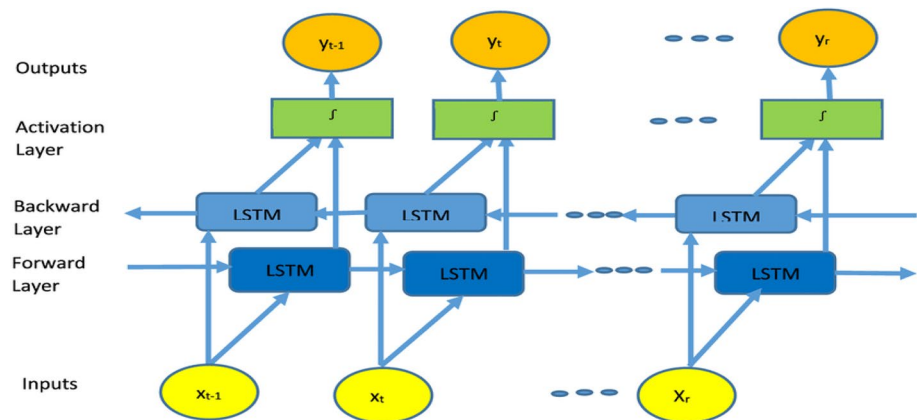
The second convolutional block has the same structure as the first, but the number of filters is increased to 64. This enables the network to extract more complicated and higher-level features from the output of the preceding layer, using the same mathematical procedures as the first convolutional block but adjusting for greater dimensionality. Table 4 represents the key parameters of the proposed hybrid deep model (CNN-BiLSTM).

3.3.7 Bi-LSTM layer: an LSTM variant that processes input sequences in both forward and reverse directions to capture comprehensive temporal dependencies

Bi-LSTM is a compelling extension of the traditional LSTM network, which captures data dependencies by processing information in both forward and backward directions [21]. In contrast to standard LSTMs, which consider only the past context [32, 33], Bi-LSTMs can understand the full context of a sequence by performing an analysis of the sequence simultaneously from the beginning to the end and vice versa [23]. This dual perspective allows the Bi-LSTMs to have much better performance on tasks that are valuable in terms of order and context, such as natural language processing,

Table 4 Significant parameter of the proposed hybrid model (CNN-BiLSTM)

Significant parameters	Values
Number of Convolutional layers	2
Number of Filters (Layer 1)	32
Kernal Size (Layer 1)	3
Activation (Layer 1)	LeakyReLU
Number of Filters (Layer 2)	64
Number of LSTM Layer	2 (Bidirectional)
LSTM Units (Layer 1)	128
LSTM Units (Layer 2)	64
Number of Dense Layer	3
Activation (Output Layer)	Softmax
Dropout	0.5
Max Pooling Size	2
Optimizer	Adam
Batch Size	128
Number of Epochs	25

Fig. 4 The general architecture of the Bi-LSTM

time series prediction, and speech recognition [27]. By including two LSTM layers, one processing the input sequence in a forward manner and the other in reverse, Bi-LSTMs can capture richer patterns with more nuances. This subsequently leads to better comprehension and prediction of sequential data [34]. The Bi-LSTM architecture is depicted in Fig. 4.

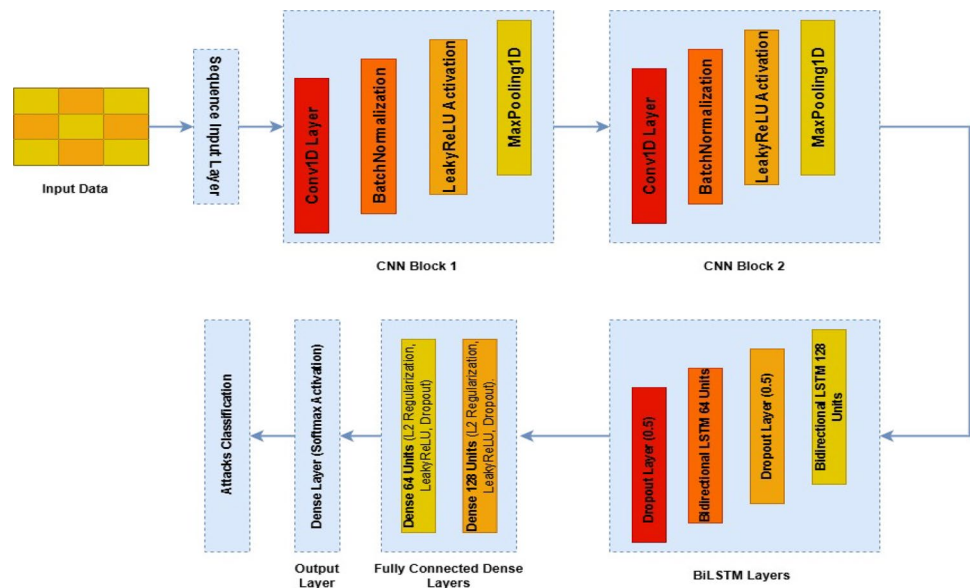
For capturing temporal dependencies in both directions, however, the Bi-LSTM layers are crucial units of the model architecture. The first Bi-LSTM layer has 128 units in each direction, meaning a total of 256 units, and is set to return the full sequence so that subsequent layers can compute over complete temporal information, which is represented in the equations below [21].

$$\text{ForwardLSTM} : \vec{h}_t = \text{LSTM}(x_t, \vec{h}_{t-1})$$

$$\text{Backward LSTM} : \overleftarrow{h}_t = \text{LSTM}(x_t, \overleftarrow{h}_{t+1})$$

$$\text{Concatation} : h_t = \text{LSTM}[\vec{h}_t, \overleftarrow{h}_t]$$

Fig. 5 Depicts the proposed model's architecture, highlighting the feature extraction layers followed by sequential processing



The operation processes the sequence through the forward and backward LSTM units, and we concatenate their outputs. A dropout layer is added, represented in the equations below,

$$h_t = \begin{cases} 0 & \text{with probability } 0.5 \\ \frac{h_t}{0.5} & \text{with probability } 0.5 \end{cases}$$

with a dropout rate of 50%, preventing overfitting by setting half of the units to zero at random during training. The second acts as a kind of sequence-to-sequence refinement with 64 units, configured otherwise to a single fixed vector length that can represent all the sequences, meaning it can be used on classification tasks. This layer is followed by a dropout layer that uses a rate of 50% for regularization. After such Bi-LSTM layers, the model proceeds with fully connected dense blocks. The first dense block is made up of 128 neurons and utilizes L2 Regularization with $\lambda = 0.001$ as a measure for overfitting due to the nature of the penalty being directly proportional to the square of weights. For nonlinearity, we want to introduce LeakyReLU, and for more reduction in overfitting, a 50% dropout is introduced. The second dense block reduces the feature representation dimensionality to 64 neurons before going to the output layer, still with the use of L2 regularization, LeakyReLU for activation, and dropout for non-linearity and regularization. Finally, a dense layer with L2 regularization on the weights serves as the output layer of the model, through which it predicts output probabilities of as many classes as in the dataset, which are represented in the below equations.

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

where p_i represents the probability of a class z_i , and C represents the number of classes.

This layer contains several neurons equal to the number of target classes. The SoftMax function normalizes the output scores and leads them to probability distributions, which are easily managed for multi-class classification. Algorithm 3 outlines the proposed hybrid deep learning (CNN-BiLSTM) model. Figure 5. depicts the proposed model's architecture, highlighting the feature extraction layers followed by sequential processing. .

i. Input Layer:

- $X \in \mathbb{R}^{n \times 1}$ be the input sequence, where n = number of features.

ii. First Convolutional Block:

- Convolutional Operation: $y_i = \sum_{k=1}^3 w_k x_{i+k-2} + b$
- Batch Normalization: $\hat{y} = \gamma \frac{y - \mu}{\sigma} + \beta$
- LeakyReLU activation:

$$f(y) = \begin{cases} y & \text{if } y > 0 \\ 0.1y & \text{if } y \leq 0 \end{cases}$$

- MaxPooling:

$$f(y) = \begin{cases} y & \text{if } y > 0 \\ 0.1y & \text{if } y \leq 0 \end{cases}$$

iii. Second Convolutional Block:

- Same operations as the first convolutional block but with 64 filters.

iv. First Bi-LSTM Layer:

- *Forward LSTM*: $\vec{h}_t = LSTM(x_t, \vec{h}_{t-1})$
- *Backward LSTM*: $\overleftarrow{h}_t = LSTM(x_t, \overleftarrow{h}_{t+1})$
- *Concatation*: $h_t = LSTM[\vec{h}_t, \overleftarrow{h}_t]$
- *Dropout*: Randomly set 50% of the activation to zero.

v. Second Bi-LSTM:

- Same operations as the first Bi-LSTM layer but without returning sequence.

vi. First Fully Connected Layer:

- Dense Layer: $z = Wx + b$
- L2 Regularization: $\text{Loss} = \frac{\lambda}{2} ||W||^2$
- LeakyReLU Activation: same as above.
- Dropout: same as above.

vii. Second Fully Connected Layer:

- Same operations as the first dense block but with 64 neurons.

viii. Output Layer:

- Dense Layer: $z = Wx + b$
- Softmax Activation:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

Where p_i represents the probability of a class z_i , and C represents the number of classes.

3.4 Evaluation criteria

Several performance metrics have been used to rigorously evaluate the effectiveness of the proposed strategy, ensuring a holistic evaluation of its capability. This involves common standard criteria that might also include accuracy, precision, recall, and F1-score, giving insight into the hybrid deep learning model's overall performance related to classification.

3.4.1 Accuracy

The percentage of accurate predictions the model makes from all forecasts is known as accuracy and is a crucial evaluation statistic. The equation below provides specifics on the accuracy formula [13, 24, 30].

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

3.4.2 Precision

It calculates the percentage of actual correctly identified instances by model. The equation below illustrates the precision calculation [13, 24, 30].

$$\text{Precision} = \frac{TP}{TP + FP}$$

3.4.3 Recall

Recall measures how well a model can identify actual instances of a class. The equation below illustrates the recall computation procedure [13, 24, 30].

$$\text{Recall} = \frac{TP}{TP + FN}$$

3.4.4 F1-score

The below equation provides the formula for calculating the F1-Score, which is the harmonic mean of recall and precision [13, 24, 30].

$$F1 - \text{Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

3.4.5 Support

Support shows how many times each class appears overall in the dataset [13, 24, 30].

3.4.6 AUC-ROC

The AUC-ROC curve is a diagnostic plot to review the performance of a classification model based on its correct and incorrect prediction rates, which are represented in Figs. 14, 15, and 16. In particular, the ROC for a classifier measures its discrimination ability, that is, the ability of a classifier to separate classes. This is done by comparing the true positive rate against the false positive rate, which indicates the sensitivity of the model. AUC stands for Area Under the Curve and provides a summary of the ROC curve [24].

3.4.7 Confusion matrix

One approach for evaluating a classification model performance is the confusion matrix represented in Figs. 11, 12, and 13. It is a table that clearly illustrates the misclassifications across several classes by summarizing the model performance on a test dataset with known true values. This matrix provides a comprehensive description of the model prediction results in a classification task, which is essential for computing several performance indicators [10, 24, 30].

Table 5 Experimental setup specification

CPU	13th Gen Intel (R) Core (TM) i9-13,900
Number of Cores	24
Number of Threads	32
Processor Frequency	2.00 GHz
Cache	36 MB L3 Cache
RAM	64 GB
Operating System	Windows 11 (64-bit)
Python Version	3.12.4
Libraries used	Keras 3.5.0, TensorFlow 2.17.0

Table 6 Classification performance of the proposed hybrid model (CNN-BiLSTM) on the KDDCup99 dataset

Attack (Class)	Precision	Recall	F1-Score
Bening (2)	100%	100%	100%
Malware (1)	92%	99%	95%
DoS (0)	100%	100%	100%
Scanning (3)	99%	100%	99%
Unauthorized (4)	41%	82%	55%
Web (5)	00%	00%	00%

In these formulas, (TP) represents True Positives, (TN) represents True Negatives, (FP) represents False Positives, and (FN) represents False Negatives. To derive the True Positive Rate (TPR) and False Positive Rate (FPR), as demonstrated in the equations below [13, 24, 30].

$$\text{True Positive Rate} = \frac{TP}{TP + FN}$$

$$\text{False Positive Rate} = \frac{FP}{TN + FP}$$

4 Empirical evaluation and results analysis

4.1 Experimental setup

It is essential to offer comprehensive details regarding the system configuration utilized throughout the study to ensure the repeatability and dependability of our studies. The proposed hybrid deep learning models were implemented using Python 3.7, leveraging the Keras and TensorFlow frameworks [1]. The experiments were run on a Dell Desktop-CFDNAIN with a 2.00 GHz processor, 64 GB of RAM, and a 64-bit Windows 11 operating system. The specific details are provided in Table 5:

4.2 Result analysis

In this section, we analyze the detailed performance of our proposed hybrid deep learning model (CNN-BiLSTM) using accuracy, validation loss, precision, recall, and F1-score and other classification metrics such as confusion matrices and Receiver Operating Characteristics (ROC) curves for different attack classes from three benchmark datasets namely KDDCup99, NSL-KDD, and CIC_IDS_2017. These metrics will be important to understand the proposed hybrid model (CNN-BiLSTM) capability regarding correct classification among the various types of IoT attacks, including benign and

Table 7 Classification performance of the hybrid model (CNN-BiLSTM) on the NSL_KDD dataset

Attack Class	Precision	Recall	F1-Score
Bening (1)	86%	92%	92%
DoS (0)	100%	100%	100%
Probe (2)	100%	100%	100%
R2L (3)	96%	98%	98%
U2R (4)	42%	71%	53%
Web (5)	5%	25%	8%

Table 8 Classification performance of the proposed hybrid model (CNN-BiLSTM) on the CIC_IDS_2017 dataset

Attack class	Precision	Recall	F1-Score
Bening (0)	100%	98%	99%
DDoS (1)	99%	100%	99%
DoS (2)	98%	100%	99%
Port Scan (3)	74%	100%	85%
Web (4)	65%	100%	79%
Bot (5)	20%	99%	34%

various kinds of attack classes. This all combines to give a complete picture of the hybrid (CNN-BiLSTM) model's overall performance on classification and the trade-off it makes on detecting actual attacks without being falsely positive.

KDDCup99 represents the classic intrusion detection problems combined with benign traffic, including attack types such as DoS, scanning, malware, web, and unauthorized access attempts. Such diversity is important for testing the hybrid model's ability to make proper distinctions between normal and malicious activity. Table 6 summarizes the performance of the hybrid models across these different classes.

The proposed hybrid model has an excellent performance on the KDDCup99 dataset, with 100% precision and recall concerning benign traffic DoS and scanning attacks. Its high accuracy shows that this proposed hybrid model can efficiently distinguish between benign and malicious traffic, without false positive or false negative instances, which is very important in realistic security systems for avoiding false alarms. With the proposed hybrid model achieving 100% precision and recall for DoS attacks, it perfectly classifies such attacks, which is key to handling the high-volume traffic emanating from such an attack. The proposed hybrid model showed a precision of 92% with a recall of 99% on malware detection, which gives an F1-score of 95%, even though it does suggest a slight precision reduction because some instances of benign were misclassified as malware. However, the high recall makes sure that almost all malware cases were correctly identified, with a very minimal chance of threats going unnoticed. Scanning attacks could be classified with almost complete accuracy: precision of 99%, and recall of 100%, with an F1-score of 99%. This indicates that the proposed hybrid (CNN-BiLSTM) model shows great proficiency in the identification of these probing activities with minimal misclassification of benign traffic.

The NSL-KDD dataset is an enhanced version of KDDCup99 to reduce redundancy and thus offers a better balance to evaluate intrusion detection systems. The following are some types of common attacks included in this dataset: DoS, Probe, R2L, Web, and U2R. Table 7 illustrates the proposed hybrid model (CNN-BiLSTM) performance on the NSL-KDD dataset.

In these, the proposed hybrid model (CNN-BiLSTM) attained a precision of 86% and a recall of 92% for benign traffic, giving an F1-score of 92%. Although the relatively lower precision compared with the KDDCup99 dataset suggests a higher number of false positives, the recall score shows that the proposed hybrid model (CNN-BiLSTM) correctly identified most of the instances of benign traffic, which is one of the key factors contributing toward the reduction in the number of false alarms in real-world scenarios. The proposed hybrid model (CNN-BiLSTM) has attained 100% precision, recall, and F1 scores for the DoS and Probe attacks. This shows the strength and robustness of the proposed hybrid model (CNN-BiLSTM) in detecting these kinds of attacks, which are predictable and highly loaded. R2L is an attack that entails unauthorized attempts at remote access; thus, the model returned to 96% precision, 98% recall, and consequently, an F1-score of 98%. This depicts the strong performance of the proposed hybrid model (CNN-BiLSTM) in identifying such generally subtle threats with very few false positives. However, User-to-Root attacks, including most of the complex privilege escalation techniques, proved more difficult, yielding 42% precision, 71% recall, and an F1 score of 53%.

The latest network traffic includes DDoS, DoS, port scanning, Web, and botnet attacks featured in the CIC_IDS_2017 dataset. Because of the larger size and diversity of data, it can be considered the best benchmark for the evaluation of

Fig. 6 Illustrates the performance of the proposed hybrid model (CNN-BiLSTM) on the KDD_Cup99 dataset, comparing training accuracy and validation accuracy

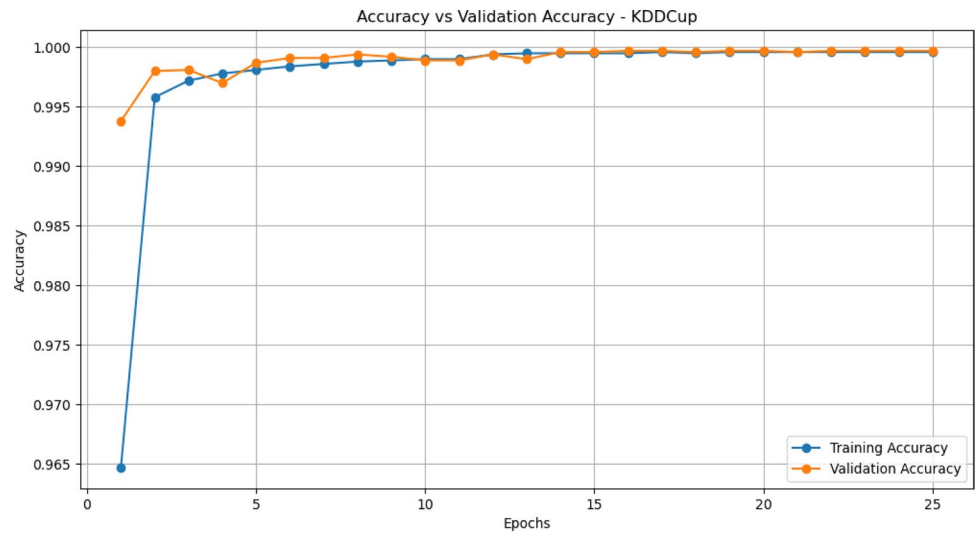


Fig. 7 Illustrate the performance of the proposed hybrid model (CNN-BiLSTM) on the NSL_KDD dataset, comparing training accuracy and validation accuracy

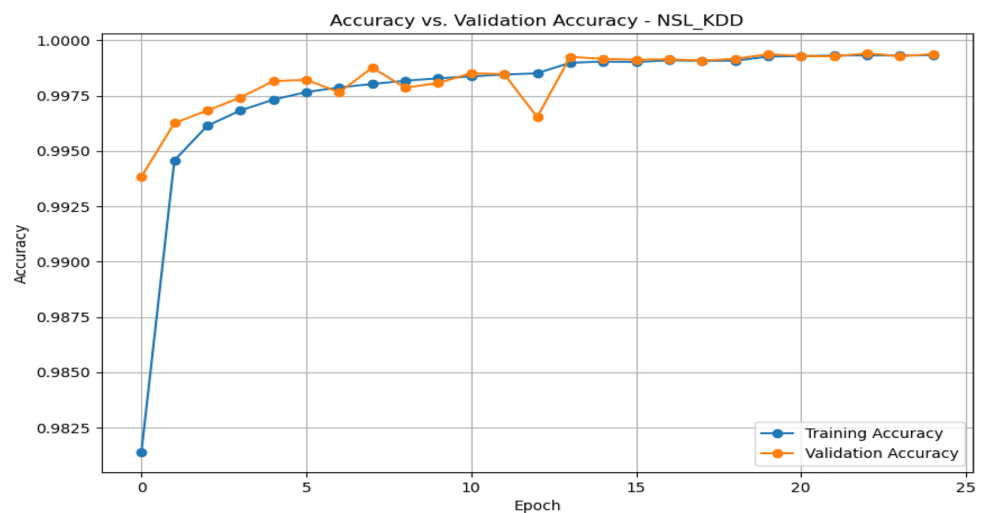


Fig. 8 Illustrates the performance of the proposed hybrid model (CNN-BiLSTM) on the CIC_IDS_2017 dataset, comparing training accuracy and validation accuracy

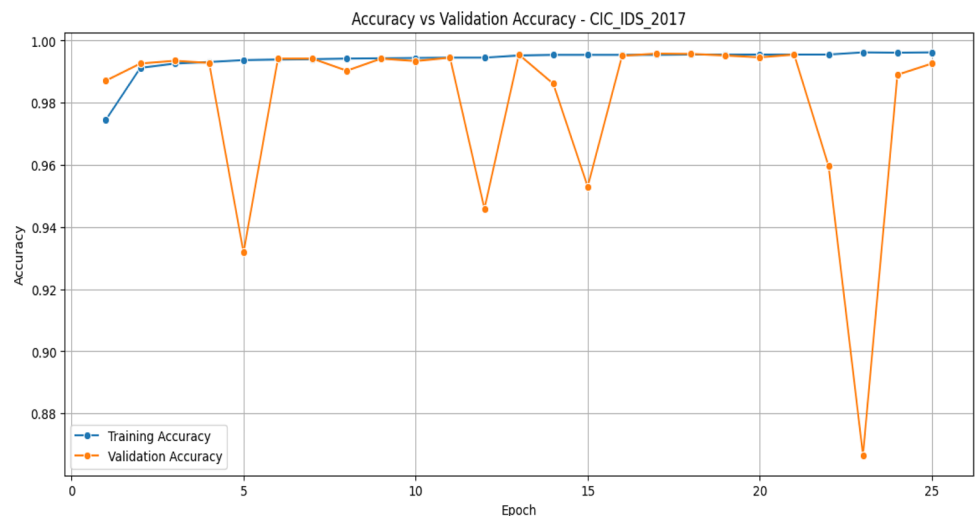


Fig. 9 Illustrates the performance of the proposed hybrid model (CNN-BiLSTM) on the KDD_Cup99 dataset, comparing loss and validation loss

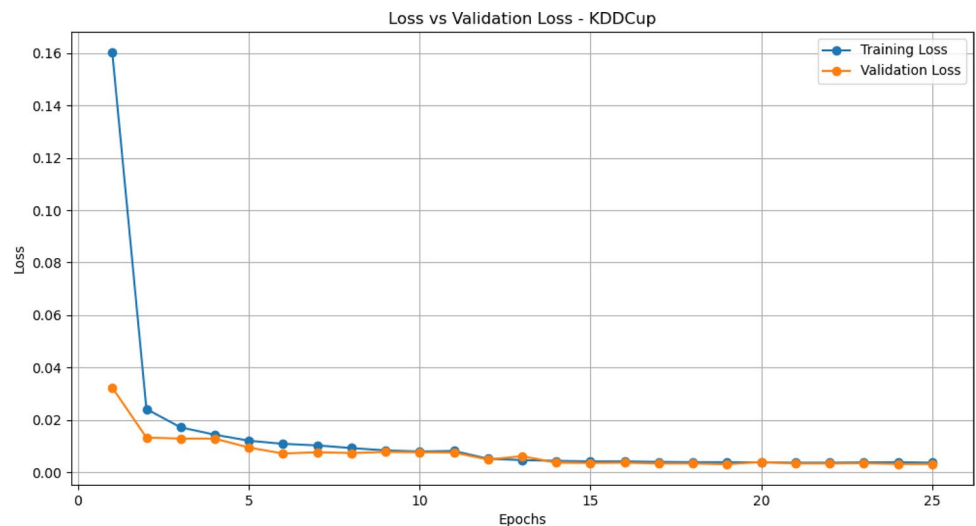
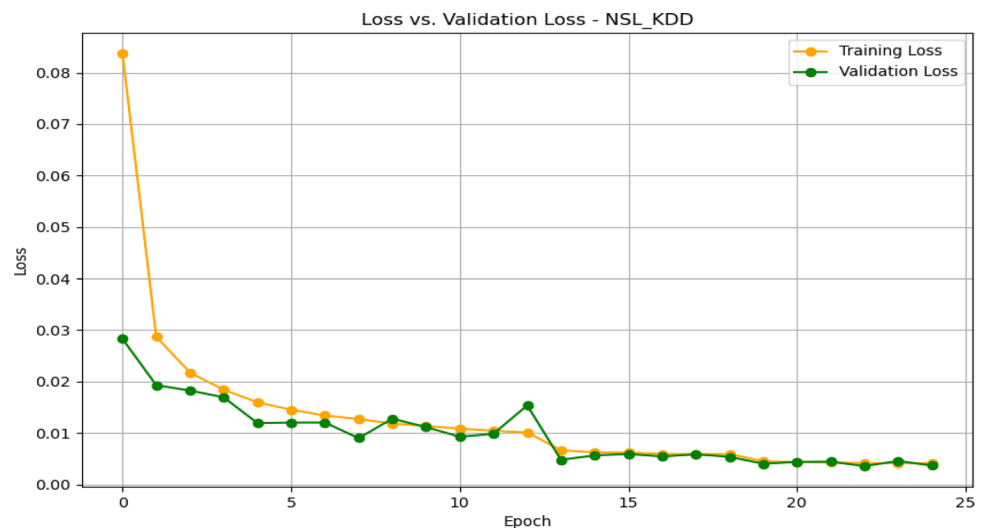


Fig. 10 Illustrates the performance of the proposed hybrid model (CNN-BiLSTM) on the NSL_KDD dataset, comparing loss and validation loss



the real-world performance of the hybrid model. Table 8 presents the hybrid model classification performance across various attack types.

Thereafter, the classification of benign traffic was done perfectly by the proposed hybrid model (CNN-BiLSTM), with a precision of 100%, resulting in a recall of 98% and an F1-score of 99%, proving its superiority in reducing the number of false positives. The proposed hybrid model (CNN-BiLSTM) performed well, especially for DDoS and DoS, performing at 99% precision, 100% recall, and an F1-score of 99%, thereby showing the high-volume attack detection capability. On the contrary, for the detection of port scanning, the proposed hybrid model (CNN-BiLSTM) showed a low performance with a precision of 74% and a recall of 100%, hence an F1-score of 85%. It detected all port scanning events; however, the lower precision mentioned that some benign traffic was misclassified, possibly due to the overlap of certain patterns in the scanning activity and normal network traffic.

The accuracy and validation accuracy across the proposed hybrid model (CNN-BiLSTM) underline its strong ability for generalization on unseen data. As shown in Figs. 6, 7, and 8, the proposed hybrid model (CNN-BiLSTM) achieved an accuracy above 99% when verified with both the KDDCup99 and NSL-KDD datasets and nearly 98% with the CIC_IDS_2017 dataset. This high performance is attributed to the proposed hybrid model (CNN-BiLSTM), which effectively captures temporal dependencies in both directions within sequences of network traffic, while the presence of convolutional layers, batch normalization, and dropout regularization helps avoid overfitting, especially for the more complicated CIC_IDS_2017 dataset.

The values of the loss are indicative of model convergence and stability. Indeed, as is easy to see from Figs. 9, 10, and 11, training loss decreased steadily during the training process. Moreover, validation loss followed a similar trend,

Fig. 11 Illustrates the performance of the proposed hybrid model (CNN-BiLSTM) on the CIC_IDS_2017 dataset, comparing loss and validation loss

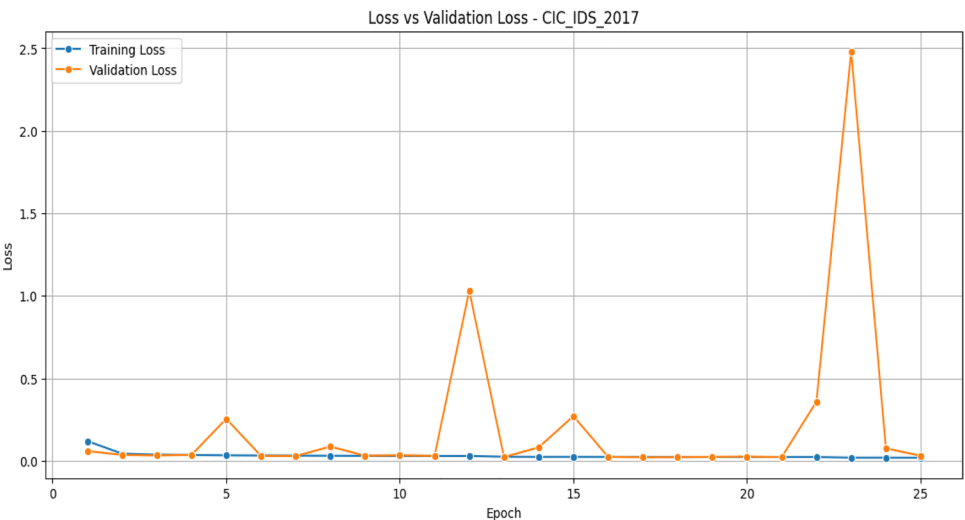


Fig. 12 Depicts the confusion matrix, illustrating the proposed hybrid model (CNN-BiLSTM) classification performance on the KDD-Cup99 dataset

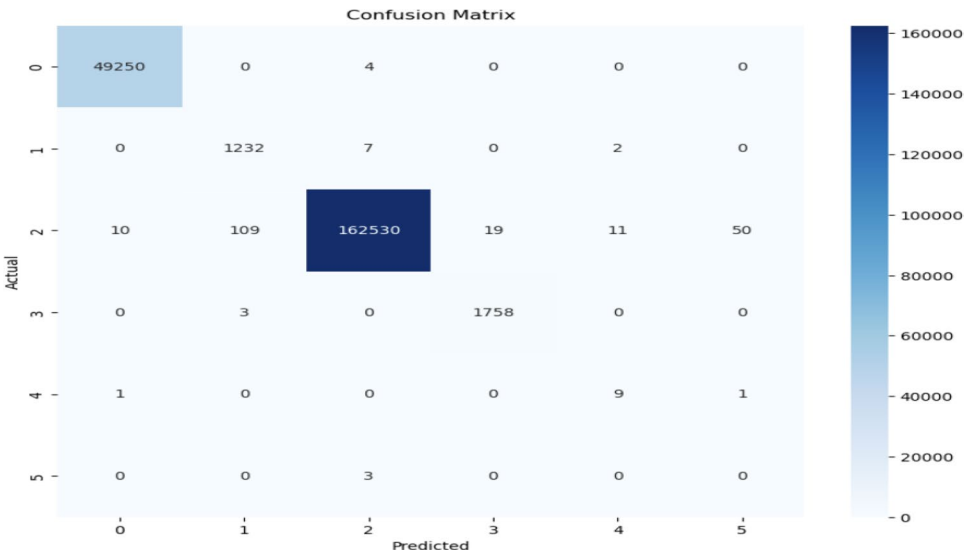


Fig. 13 Depicts the confusion matrix, illustrating the proposed hybrid model (CNN-BiLSTM) classification performance on the NSL_KDD dataset

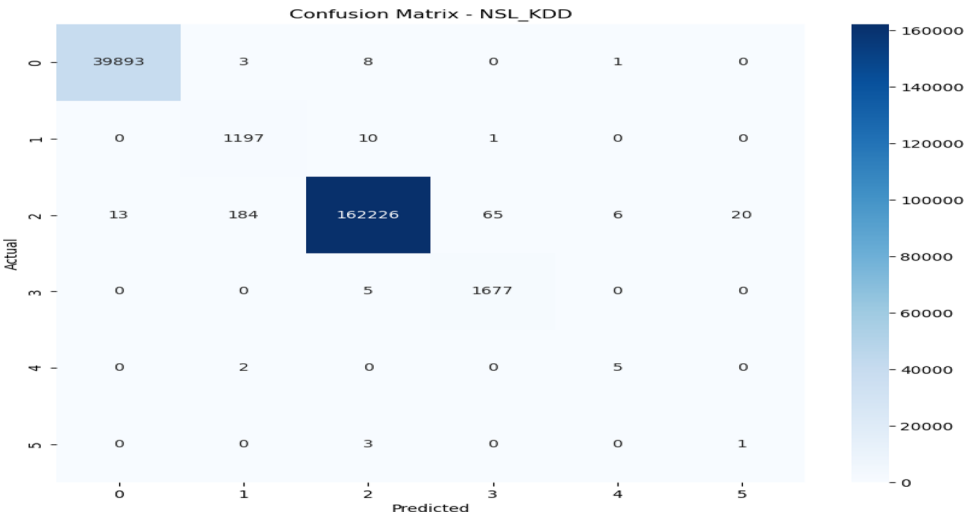


Fig. 14 Depicts the confusion matrix, illustrating the proposed hybrid model (CNN-BiLSTM) classification performance on the CIC_IDS_2017 dataset

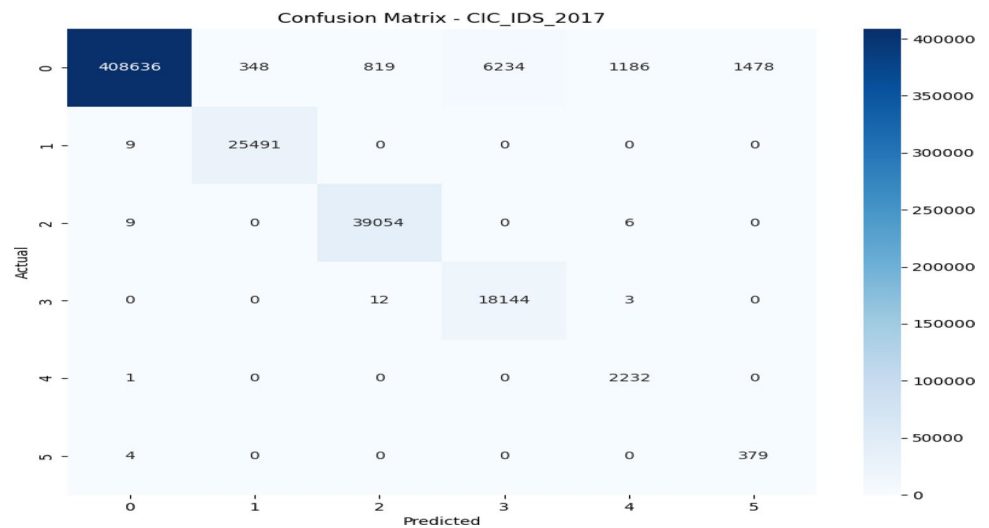


Fig. 15 Illustrates the ROC curve for the KDDCup99 dataset

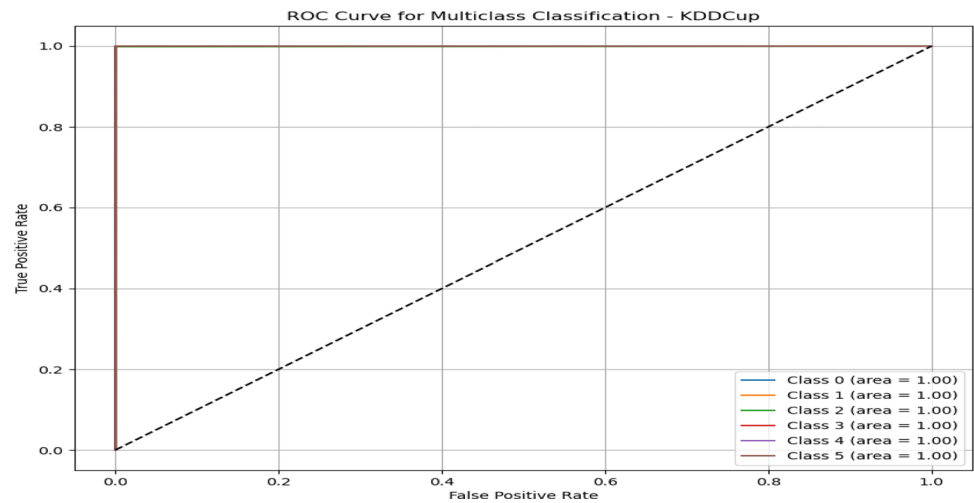


Fig. 16 Illustrates the ROC curve for the NSL_KDD dataset

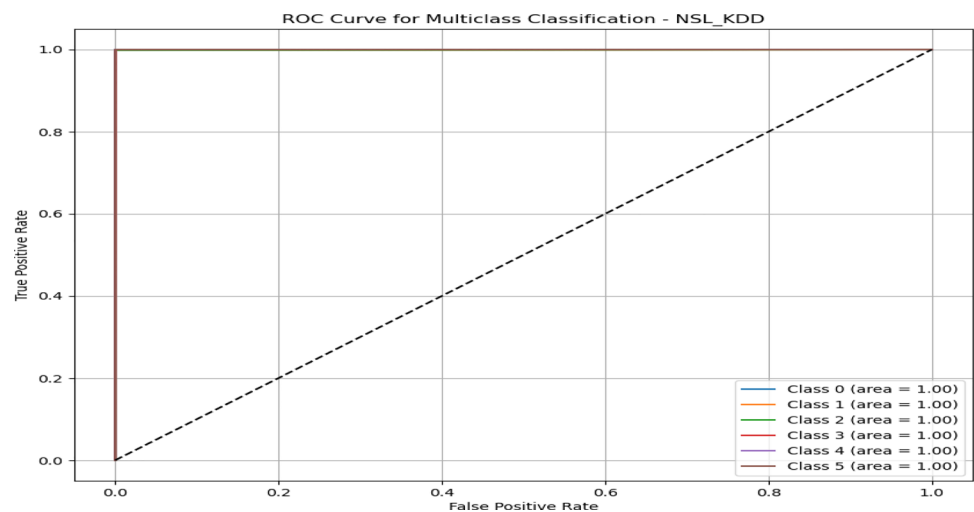


Fig. 17 Illustrates the ROC curve for the CIC_IDS_2017 dataset

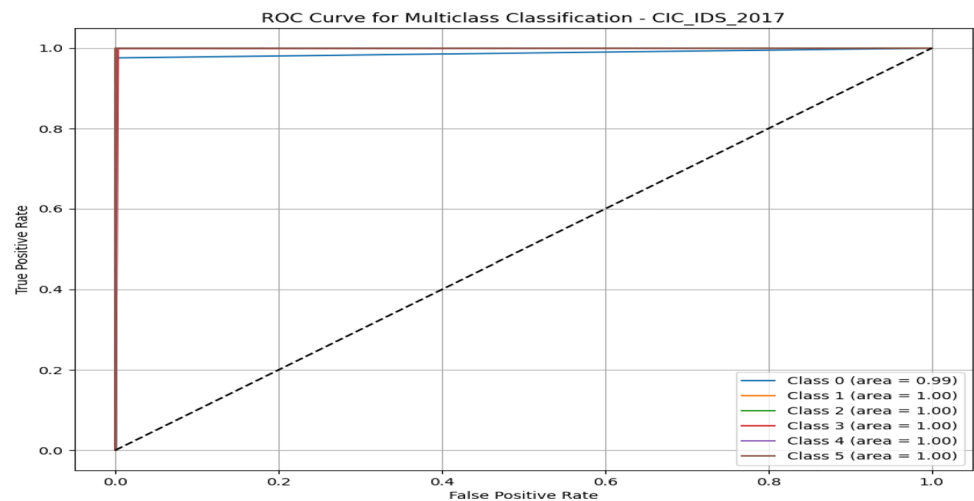
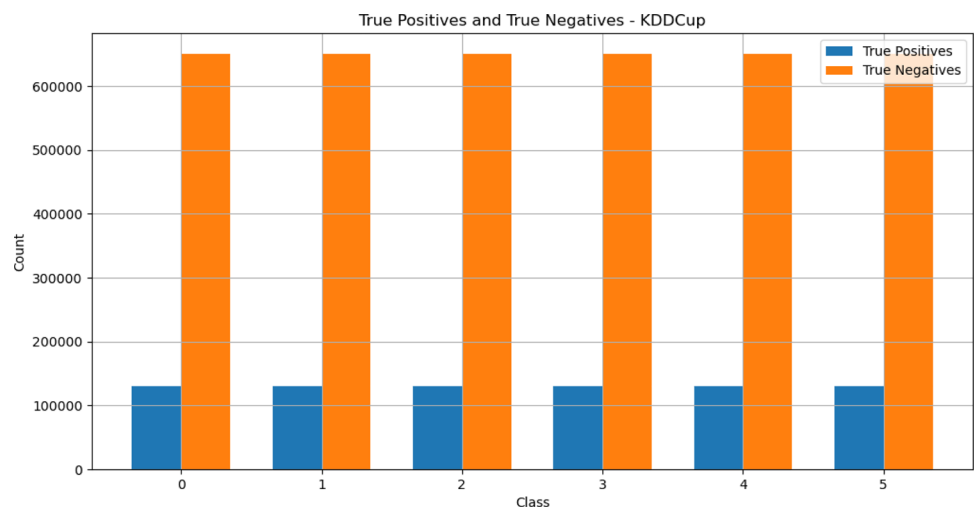


Fig. 18 Illustrates the true positive and true negative rates for the KDDCup99 dataset



resulting in good generalization. It has low validation loss for all three datasets. Its performance in KDDCup99 is even stronger compared to CIC_IDS_2017 and NSL-KDD, with even lower loss values. That will indicate that the proposed hybrid model (CNN-BiLSTM) learned both modern and traditional network traffic attack patterns well and was able to maintain its robustness against different types of variations in the case of an IoT attack.

The confusion matrices for all three datasets, CIC_IDS_2017, KDDCup99, and NSL-KDD-present a clear picture of the proposed hybrid model (CNN-BiLSTM) performance regarding classification, in Figs. 12, 13, and 14. The confusion matrices present True Positives, True Negatives, False Positives, and False Negatives. The model performed very well, classifying nearly all benign and attack traffic with very low rates of FP and FN, which are good IoT attack detections.

The ROC curves of each dataset are shown in Figs. 15, 16, and 17, along with their respective AUC scores to further analyze the performance of our proposed hybrid model (CNN-BiLSTM). A ROC curve represents a combination of sensitivity (True Positive Rate) and specificity (False Positive Rate) at different classification thresholds. Indeed, on the CIC_IDS_2017 dataset, our classifier gave a very outstanding AUC score of 0.99, showing a near complete separation between classes. The corresponding AUC values obtained over the KDDCup99 and NSL-KDD datasets are 100 and 100, respectively, reflecting the robustness of the proposed hybrid model (CNN-BiLSTM) in terms of effectively distinguishing between normal and attack traffic in diverse IoT attack detection scenarios.

Figures 18, 19, and 20 show the distribution of true positives and true negatives between different classes of attacks for KDDCup99, NSL-KDD, and CIC_IDS_2017 datasets, respectively. The proposed hybrid model (CNN-BiLSTM) identifies a high number of true negatives among all classes across all three datasets. This reflects its robustness for correctly identifying benign traffic. True positives vary depending on the type of attack. The proposed hybrid model

Fig. 19 Illustrates the true positive and true negative rates for the NSL_KDD dataset

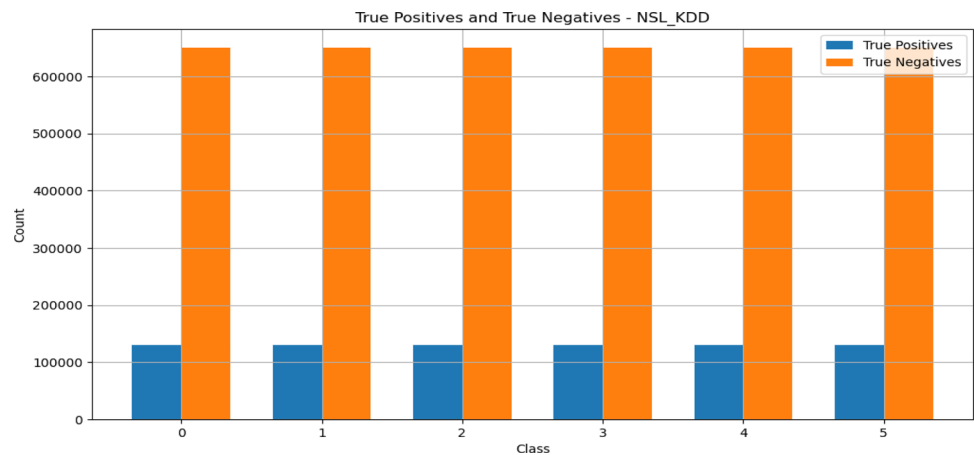
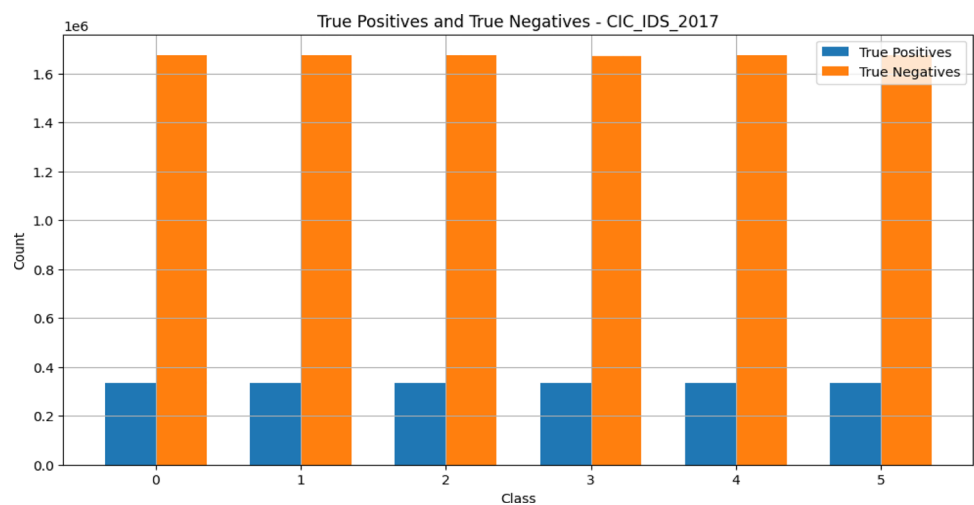


Fig. 20 Illustrates the true positive and true negative rates for the CIC_IDS_2017 dataset



(CNN-BiLSTM) gives a very high count of true positives for well-represented classes like DoS and DDoS, hence high performance in these cases.

Our proposed approach with (CNN + Bi-LSTM) outperforms many state-of-the-art models as shown in Table 9, on different datasets, such as KDDCup99, NSL-KDD, and CIC_IDS_2017, with higher accuracy, precision, recall, and F1-scores. For example, [1] applied RBM, CNN, and GAP techniques on smart city datasets which obtained an accuracy of around 98%. As per our proposed hybrid model (CNN-BiLSTM), KDDCup99 records an accuracy of 99.90%, while its precision, recall, and F1 also record values of 99.90%, hence showing a well-balanced performance in these metrics. Similarly, [3] presented an accuracy of 99.40% and across all metrics, 99% in DS2OS using Gradient Boosting. However, our proposed hybrid model (CNN-BiLSTM) outperforms the latter in both recall and F1, those metrics being critical for ensuring minimal false negatives. The CNN model proposed in [9] had an accuracy of 99.75% for DDoS on CICDDoS2019, but it is lacking in precision, recall, and F1-scores, whereas our proposed hybrid model (CNN-BiLSTM) was able to achieve 97.97% on CIC_IDS_2017 with all the key metrics at 99.90%. Proposed the model using MLP, RNN, and DNN on different attacks, which further showed an accuracy of 96.90%, precision of 95.21%, recall of 94.50%, and F1 of 95.03%, though the model we proposed outperformed it over NSL-KDD with an accuracy of 99.84%, precision of 96.00%, recall of 99.90%, and F1 of 98.00%. In addition, the work in [34] applied CNN + LSTM on IoT datasets, showing an accuracy of 94.32% and an F1-score of 98.60%, but our model outperformed it by attaining an accuracy of 97.97% and in all metrics by 99.90% on CIC_IDS_2017. In the last turn, CNNA-BiLSTM from [21] secured the accuracy on InSDN of 99.31%, but our approach shows superior results in recall and F1-score. By and large, comprehensive performance is outstanding for our proposed hybrid model (CNN-BiLSTM) within the whole range of detection of IoT attack types, with few false positives and negatives, which confirms efficiency and state-of-the-art status in IoT attack detection.

Table 9 Presents a comparison of performance metrics showing the advantages of our hybrid approach over existing models

Author	Techniques	Types of attacks	Dataset	Results: AC (accuracy), PR (precision), RE (recall), F1-score
[1]	RBM, CNN, GAP	DDoS, Replay	Smart city datasets (environmental, smart river, and smart soil	AC on environment dataset: 98.37% AC on river dataset: 98.13% AC on soil dataset: 99.51%
[3]	Gradient Boosting	DoS, DP, MC, MO, Scan, Spying, Wrong Setup	DS2OS	AC: 99.40% PR: 99% RE: 99% F1-score: 99%
[9]	CNN	DDoS	CICDDoS2019	AC: 99.75 PR: Not mentioned RE: Not mentioned F1-score: Not mentioned
[15]	MLP, RNN, DNN	BruteForceXXS, BruteForceWEB, DoS_Hulk_Attack, and DOS_LOIC_HTTP_Attack	NSL-KDD, KDDCup99, UNSW-NB15	AC: 96.90% PR: 95.21% RE: 94.50% F1-score: 95.03%
[17]	DNN, DT	NMRI, CMRI, MSCI, MPCl, MFCI, DoS, Recon	Real-world ICS Gas dataset, Secure Water Treatment (SWaT)	AC: 96.20% PR: 96.17% RE: 96.20% F1-score: 96.18%
[11]	CNN	DDoS	NSL-KDD	AC: 92.99% PR: Not mentioned RE: Not mentioned F1-score: Not mentioned
[12]	SDAE-SVM	IoT attacks	KDD-CUP99	AC: 98.12% PR: Not mentioned RE: Not mentioned F1-score: Not mentioned
[14]	Constant removal and Recursive Feature Elimination with SVM, DT, and RF	Injection attack	AWID	AC: 99% PR: 95.39% RE: 85.00% F1-score: 91.58%
[34]	CNN + LSTM	MitM, Intrusion, DoS	IoT-23, N-BaloT, and CICIDS2017	AC: 94.32% PR: 99.32% RE: 97.88% F1-score: 98.60%
[21]	Neural Network with Attention (CNNA)-BiLSTM	Probe, DoS, web attacks, Brute Force, Botnet, U2R, and DDoS	InSDN	AC: 99.31% PR: 99.32% RE: 99.31% F1-score: 99.31%

Table 9 (continued)

Author	Techniques	Types of attacks	Dataset	Results: AC (accuracy), PR (precision), RE (recall), F1-score
[23]	RNN-BiLSTM-CRF	Theft attack	State Grid Corporation of China (SGCC), Local dataset, and Electricity on Australia's East Coast (AUSGRID)	AC: 93.05% PR: 92.02% RE: 84.18% F1-score: 86.62%
[30]	CNN + LSTM	BASHLITE and Mirai	N-BaloT	PR: 94% RE: 87% F1-score: 94%
Proposed	LASSO, SMOTE, and CNN + Bi-LSTM	DDoS, DoS, Probe, R2L, U2R, Web, Malware, Scanning, Unauthorized, Bot, and Port Scan	KDDCup99	AC: 99.90% PR: 99.90% RE: 99.90% F1-score: 99.90%
			NSL-KDD	AC: 99.84% PR: 96.00% RE: 99.90% F1-score: 98.00%
			CIC_IDS_2017	AC: 97.97% PR: 99.90% RE: 99.90% F1-score: 99.90%

Fig. 21 Space Complexity Analysis

Space Complexity Analysis

Metric	Value
Model Parameters	393,670
Memory Usage (Approximate)	1.50 MB

Fig. 22 Time Complexity Analysis

Time Complexity Analysis

Component	Time Complexity
Feature Selection (LassoCV)	$O(n^2)$
SMOTE Resampling	$O(n \log n)$
CNN-LSTM Training	$O(n * m * d)$
Inference Time	$O(d)$

The findings of our proposed CNN-BiLSTM model demonstrate superior performance across multiple benchmark datasets when compared with existing approaches. For instance, on the KDDCup99 dataset, our model achieved 99.90% in accuracy, precision, recall, and F1-score, outperforming SDAE-SVM [12], which reported only 98.12% accuracy with other metrics not mentioned. Similarly, on the NSL-KDD dataset, our model's accuracy of 99.84% significantly exceeds that of CNN-based approaches [11], which achieved only 92.99% accuracy without reporting precision, recall, or F1-score. Even when compared to the advanced CNNA-BiLSTM model [21], which achieved an overall performance of 99.31% across all metrics on the In SDN dataset, our model shows higher accuracy and precision, especially notable in detecting complex attacks.

On the CIC_IDS_2017 dataset, our model achieved an accuracy of 97.97% with precision, recall, and F1-score, all reaching 99.90%, which is substantially higher than the CNN + LSTM model [34] that achieved only 94.32% accuracy and the CNN-only model [9], which reported 99.75% accuracy but omitted other evaluation metrics. Moreover, while [14] reported high accuracy (99%) on the AWID dataset, their recall (85%) and F1-score (91.58%) reveal a performance gap, particularly in handling imbalanced or complex attack classes.

One notable inconsistency is seen in models like [23] (RNN-BiLSTM-CRF), where despite the use of complex architectures, the recall drops significantly (84.18%), indicating difficulty in detecting certain attack types. Our model, however, maintains consistently high recall across diverse attacks, including DoS, DDoS, Probe, R2L, and U2R, suggesting better generalizability and robustness.

Overall, our model not only outperforms existing works in terms of average metrics across datasets but also demonstrates stability and reliability across a wider range of attack types, making it more applicable to real-time IoT environments. Nonetheless, further research should continue to examine why some methods fail in specific attack scenarios, particularly where recall drops or datasets are imbalanced, and how hybrid models like ours can mitigate such issues.

We note that some earlier studies have utilized other datasets that may bear different characteristics from the ones considered in this article, thus making a direct comparison of results not possible. However, the objective of our research is benchmarking relative to these widely established datasets, KDDCup99, NSL-KDD, and CIC_IDS_2017, which are commonly adopted in various intrusion detection-related research. A closer look at Table 8 reveals that other works, such as [13, 29], and [10], had also used benchmarks with KDDCup99 and NSL-KDD, making them relevant benchmarks to our evaluation. Besides, the preprocessing techniques feature selection using LASSO and balancing by SMOTE, as well as the model architecture CNN-BiLSTM, strongly indicate performance improvement. The gains in accuracy, precision, recall, and F1-score in several datasets point out that our approach indeed enhances generalizability and robustness.

The space and time complexity analysis of the provided model in Figs. 21 and 22 addresses concerns about real-time feasibility. The Space Complexity Analysis shows that this model has 393,670 parameters and 1.50 MB memory usage, making this a relatively light model for environments with resource constraints. The Time Complexity Analysis further specializes in the computational requirements for each stage into feature selection ($O(n^2)$), SMOTE resampling ($O(n \log n)$), CNN-BiLSTM training ($O(n * m * d)$), and inference time ($O(d)$). The inference time complexity is $O(d)$, meaning that if it can be trained once, it would allow for fast prediction and thus real-time detection claims. The proposed model deployment strategy is aligned with its real-time detection and low computational costs. The model, being lightweight and at 1.50 MB of memory usage and an $O(d)$ time complexity for inference, is more aligned with edge layer deployment or on IoT components themselves. It deploys the balance between computational power, thereby bringing near-low-latency inference at the edge, accompanied by the minimization of the dependency on cloud infrastructure while ensuring real-time threat detection. Further, edge deployment minimizes data transmission overhead, enhancing privacy and security by processing data closer to the source. However, cloud deployment is feasible for large-scale analytics as well

as model retraining periodically. The final choice depends on the specific use case, hardware constraints, and security considerations, which should be explored in future work.

5 Conclusion and future directions

In conclusion, the proposed hybrid CNN-BiLSTM model demonstrated exceptional performance in detecting high-volume and complex IoT attacks, achieving impressive accuracy rates of 99.9% on KDDCup99, 99.8% on NSL-KDD, and 98.0% on CIC_IDS_2017. The model consistently delivered high precision, recall, and F1-scores across all attack categories, particularly for sophisticated threats such as DoS, DDoS, and malware. Comparative analysis with existing state-of-the-art methods highlights the model's superior detection capability, making it highly suitable for real-time threat detection in smart IoT environments. The model's ability to effectively capture both spatial and temporal features of network traffic underscores the advantages of hybrid deep learning approaches. Designed to enhance the security of IoT platforms by accurately identifying severe and frequent attacks, the model proves promising for practical applications. However, to address limitations related to rare or complex attack types, future work will explore advanced techniques such as transfer learning and adversarial training. Additionally, ensuring scalability and efficiency for deployment in large-scale, real-time IoT environments remains a critical direction for further research and development. The proposed model can be deployed in smart homes, industrial control systems, and healthcare IoT settings to provide early warning signals against intrusions. By integrating the model with existing network infrastructure and updating it continuously, real-time and adaptive security can be achieved without human intervention.

6 Future recommendations

Cybersecurity threats in IoT environments are constantly shifting, which calls for persistent research and improvements in intrusion detection systems. Even though our proposed hybrid (CNN-BiLSTM) model showcases high accuracy and efficiency, future work should pay more attention to improving its robustness, scalability, and real-world applicability. First, testing the model on IoT-specific datasets such as TON_IoT, Bot-IoT, or IoT-23 would guarantee its effectiveness in a real-world IoT attack scenario. Second, optimization of the model for resource-constrained IoT devices through techniques such as quantization, pruning, federated learning, and hardware acceleration on AI chips (e.g., TPUs) would enhance deployment feasibility. Third, adversarial machine learning threats could be addressed through adversarial retraining, GAN-generated attack traffic, and explainable AI (XAI) techniques to improve resilience against sophisticated cyber threats. Fourth, implementation in real-world IoT security platforms would require adaptive learning, automated feature engineering, and real-time response mechanisms for the effective mitigation of emerging cyber threats. Lastly, the ability to expand from network traffic analysis into multi-modal threat detection would further strengthen the frameworks of IoT security by incorporating both behavioral analysis of IoT devices and log-based security monitoring.

Author contributions TSB: Resources, Supervision, Validation, Writing-review, and Editing. MZ: Conceptualization, Data curation, Investigation, Methodology, Software, visualization, writing original draft, writing-review and editing.

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Data availability For my manuscript, I have utilized three datasets: KDDCup99, NSL-KDD, and CIC_IDS_2017. Below is the DAS statement I will include in the system: "The datasets used and analyzed during the current study are publicly available: KDDCup99 dataset is available in the UCI Machine Learning Repository. NSL-KDD dataset is available in the University of New Brunswick Repository. CIC_IDS_2017 dataset is available in the Canadian Institute for Cybersecurity Repository." If any additional information is required, or if adjustments to the statement are necessary, kindly let me know. I will ensure the DAS is submitted correctly through the system.

Declarations

Ethics approval and consent to participate Not applicable.

Informed consent Not applicable.

Competing interests The authors declare no competing interests.

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